

CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 4 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

Q1. A mixture of nitric acid and water boils at a temperature higher than the boiling points of both pure nitric acid and pure water. This indicates that the solution: (a) Shows positive deviation from Raoult's law (b) Shows negative deviation from Raoult's law (c) Forms a minimum boiling azeotrope (d) Behaves as an ideal solution [Solutions] [PYQ 2019] [Confidence: Very High]

Q2. The correct unit of molar conductivity (Λ_m) is: (a) $S\ cm^{-1}$ (b) $S\ cm^2\ mol^{-1}$ (c) $S\ cm^{-2}\ mol$ (d) $S\ cm\ mol^{-1}$ [Electrochemistry] [PYQ 2014] [Confidence: Very High]

Q3. For the photochemical reaction $H_2(g) + Cl_2(g) \xrightarrow{h\nu} 2HCl(g)$, the order of the reaction is: (a) Zero (b) First (c) Second (d) Fractional [Chemical Kinetics] [PYQ 2016] [Confidence: Very High]

Q4. The highest oxidation state of Manganese (Mn) in its fluorides is +4 (MnF_4), but in its oxides, it is +7 (Mn_2O_7). The reason for this is: (a) Fluorine is more electronegative than oxygen. (b) Fluorine does not possess d-orbitals. (c) Oxygen can form multiple bonds (double bonds) with transition metals. (d) Fluorine atoms are larger in size than oxygen atoms. [The d- and f-Block Elements] [PYQ 2016] [Confidence: Very High]

Q5. According to Valence Bond Theory (VBT), the hybridization of the central metal atom in the complex $[Ni(CO)_4]$ is: (a) dsp^2 (b) sp^3 (c) d^2sp^3 (d) sp^3d^2 [Coordination Compounds] [PYQ 2015] [Confidence: Very High]

Q6. The major product formed when 2-bromopentane is heated with alcoholic KOH is: (a) Pentan-2-ol (b) Pent-1-ene (c) Pent-2-ene (d) Pentane [Haloalkanes and Haloarenes] [PYQ Term-1 2021] [Confidence: Very High]

Q7. Which of the following reagents is the best choice to convert phenol into 2,4,6-tribromophenol? (a) Br_2 in CS_2 at $273\ K$ (b) Br_2 in CH_3COOH (c) Bromine water ($Br_2(aq)$) (d) HBr [Alcohols, Phenols and Ethers] [PYQ 2020] [Confidence: High]

Q8. When benzaldehyde is heated with concentrated $NaOH$, the products formed are: (a) Benzyl alcohol and Phenol (b) Benzyl alcohol and Sodium benzoate (c) Benzoic acid and Benzene (d) Sodium benzoate and Phenol [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2021] [Confidence: Very High]

Q9. The value of the van't Hoff factor (i) for a dilute aqueous solution of potassium hexacyanidoferrate(II), $K_4[Fe(CN)_6]$, assuming complete dissociation, is: (a) 2 (b) 3 (c) 4 (d) 5 [Solutions] [Practice] [Confidence: High]

Q10. In the plot of molar conductivity (Λ_m) versus the square root of concentration ($\sqrt{c}$) for a strong electrolyte, the slope of the straight line is: (a) Positive (b) Zero (c) Negative (d) Exponential [Electrochemistry] [PYQ 2019] [Confidence: High]

Q11. The role of a catalyst in a chemical reaction is to: (a) Increase the equilibrium constant (b) Lower the activation energy (c) Decrease the enthalpy of reaction (d) Increase the activation energy [Chemical Kinetics] [PYQ 2017] [Confidence: Very High]

Q12. The best reagent to convert primary alcohols into their corresponding alkyl chlorides without the formation of any side liquid products is: (a) PCl_5 (b) PCl_3 (c) $HCl/ZnCl_2$ (d) $SOCl_2$ [Haloalkanes and Haloarenes] [PYQ 2019] [Confidence: Very High]

Directions for Q13 to Q16: A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): Cu^{2+} is much more stable than Cu^+ in an aqueous solution. Reason (R): The high hydration enthalpy of Cu^{2+} in water heavily compensates for its high second ionization enthalpy. [The d- and f-Block Elements] [PYQ 2020] [Confidence: Very High]

Q14. Assertion (A): The complex $[Co(NH_3)_6]^{3+}$ is an inner orbital complex. Reason (R): Ammonia (NH_3) acts as a strong field ligand for Co^{3+} , forcing the pairing of $3d$ electrons and leaving inner $3d$ orbitals empty for hybridization. [Coordination Compounds] [PYQ 2015] [Confidence: Very High]

Q15. Assertion (A): The boiling point of ethanol is significantly higher than that of its isomer, dimethyl ether. Reason (R): Ethanol molecules undergo strong intermolecular hydrogen bonding, whereas dimethyl ether molecules do not. [Alcohols, Phenols and Ethers] [PYQ 2015] [Confidence: Very High]

Q16. Assertion (A): The oxidation of ketones is much easier than the oxidation of aldehydes. Reason (R): Ketones lack a hydrogen atom directly attached to the carbonyl carbon, meaning their oxidation requires breaking strong carbon-carbon bonds. [Aldehydes, Ketones and Carboxylic Acids] [Practice] [Confidence: High]

Section B: 2-Mark Questions (Short Answer)

Q17. State Raoult's law for a solution of two volatile liquids. Write the thermodynamic conditions (ΔH_{mix} and ΔV_{mix}) for an ideal solution. [Solutions] [PYQ 2019] [Confidence: Very High]

Q18. State Faraday's First Law of Electrolysis. Calculate the amount of charge (in Faradays) required for the complete reduction of 1 mole of Al^{3+} ions to Al . [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q19. Give reasons for the following: (i) Transition elements exhibit variable oxidation states. (ii) Zinc ($Z = 30$) is generally not considered a transition element. [The d- and f-Block Elements] [PYQ 2014] [Confidence: Very High]

Q20. Account for the following observations: (i) Grignard reagents ($RMgX$) must be prepared strictly under anhydrous conditions. (ii) Alkyl halides, despite being highly polar, are completely immiscible (insoluble) in water. [Haloalkanes and Haloarenes] [PYQ 2017] [Confidence: Very High]

Q21. Write the balanced chemical equation for the hydroboration-oxidation of propene. Name the major alcohol product formed. [Alcohols, Phenols and Ethers] [PYQ 2016] [Confidence: High]

Section C: 3-Mark Questions (Short Answer)

Q22. A solution containing **15 g** of urea (molar mass = **60 g mol⁻¹**) dissolved per litre of solution in water has exactly the same osmotic pressure as a solution of glucose (molar mass = **180 g mol⁻¹**) in water at the same temperature. Calculate the mass of glucose present in one litre of its solution. [Solutions] [PYQ 2018] [Confidence: Very High]

Q23. A first-order reaction has a specific rate constant of **1.15 × 10⁻³ s⁻¹**. How long will it take for **5 g** of this reactant to reduce to exactly **3 g**? (Given: **log 5 = 0.6990**, **log 3 = 0.4771**). [Chemical Kinetics] [PYQ 2018] [Confidence: Very High]

Q24. Calculate the standard Gibbs free energy change (ΔG°) for the following reaction at **298 K**:
 $Zn(s) + Ag_2O(s) + H_2O(l) \rightarrow Zn^{2+}(aq) + 2Ag(s) + 2OH^-(aq)$ Given that the standard cell potential (E°_{cell}) for this reaction is **1.56 V**. (**1F = 96500 C mol⁻¹**). [Electrochemistry] [Practice] [Confidence: High]

Q25. Assign suitable reasons for the following observations in d-block and f-block chemistry: (i) The Lanthanoid contraction is slightly less pronounced from element to element than the Actinoid contraction. (ii) Vanadium pentoxide (V_2O_5) acts as an excellent industrial catalyst. (iii) Cr^{2+} is a strong reducing agent despite having a d^4 configuration. [The d- and f-Block Elements] [PYQ 2023 / 2020] [Confidence: Very High]

Q26. (i) Draw the geometrical isomers (cis and trans) of the complex ion $[Co(NH_3)_4Cl_2]^+$. (ii) Which of these two isomers possesses a non-zero dipole moment? (iii) Explain the concept of linkage isomerism with a suitable coordination compound example. [Coordination Compounds] [PYQ 2016] [Confidence: Very High]

Q27. Write the chemical equations to show how you will convert the following: (i) Phenol to Anisole (ii) Propan-2-ol to 2-Methylpropan-2-ol (iii) Cumene to Phenol [Alcohols, Phenols and Ethers] [PYQ 2015 / Practice] [Confidence: High]

Q28. Predict the specific structures of the major organic products formed in the following reactions: (i) Propanone is treated with hydroxylamine (NH_2OH) in a weakly acidic medium. (ii) Benzaldehyde reacts with phenylhydrazine ($H_2N - NH - C_6H_5$) in an acidic medium. (iii) Ethanal reacts with Hydrogen cyanide (HCN). [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015 / CBSE SQP 2022] [Confidence: Very High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Kinetics of Radioactive Decay Chemical kinetics not only governs laboratory reactions but also natural phenomena. All natural and artificial radioactive decays of unstable nuclei follow first-order kinetics. For instance, Carbon-14 dating, which is heavily used in archaeology to determine the age of ancient biological artifacts, relies entirely on the steady first-order decay rate of the **C – 14** isotope. The time taken for half of the initial concentration of a

reactant to decay is called its half-life, which for a first-order process is entirely independent of the starting amount. (a) Define the term 'half-life' ($t_{1/2}$) of a reaction. (1 Mark) (b) Write the mathematical relationship between the half-life ($t_{1/2}$) and the decay constant (k) for a first-order reaction. (1 Mark) (c) The half-life for the radioactive decay of $C-14$ is 5730 years. If a freshly excavated archaeological wooden sample contains only 25% of the $C-14$ found in a living tree, calculate the approximate age of the artifact. (2 Marks) [Chemical Kinetics] [Practice based on PYQ Themes] [Confidence: High]

Q30. Case Study: Polyhalogen Compounds and the Environment Carbon compounds containing more than one halogen atom are known as polyhalogen compounds. Many of these are highly useful in industry and agriculture. For example, Chloroform is a solvent, and Freons (chlorofluorocarbons) are widely used as refrigerants. However, environmental concerns have heavily restricted their use. Freons easily diffuse into the stratosphere, where they initiate radical chain reactions that deplete the protective ozone layer. Similarly, DDT, once a widely used insecticide, was banned due to its high chemical stability, lack of biodegradability, and biomagnification in the food chain. (a) Write the exact IUPAC name of the insecticide DDT. (1 Mark) (b) Chemically, why are Freons (CFCs) considered extremely harmful to the environment? (1 Mark) (c) Write the balanced chemical reaction for the preparation of chloroform ($CHCl_3$) from ethanol using bleaching powder (or state the overall Haloform reaction). Alternatively, write the chemical formula of Freon-12. (2 Marks) [Haloalkanes and Haloarenes] [PYQ Theme 2016 / 2019] [Confidence: Medium]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) Define molar conductivity (Λ_m). How does the molar conductivity of a weak electrolyte vary with the decrease in concentration (dilution)? (2 Marks) (b) The conductivity of a $0.001028 \text{ mol L}^{-1}$ solution of acetic acid is $4.95 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its dissociation constant (K_a) if the limiting molar conductivity (Λ_m°) for acetic acid is given as $390.5 \text{ S cm}^2 \text{ mol}^{-1}$. (3 Marks) [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q32. (a) Using Valence Bond Theory (VBT), deduce the hybridization, geometry, and magnetic properties of the complex ion $[CoF_6]^{3-}$. (Given: Atomic number of Co = 27). (3 Marks) (b) Write the exact IUPAC name of the coordination complex $[Pt(NH_3)_2Cl(NO_2)]$. (1 Mark) (c) State one biological importance of coordination compounds, giving a specific example. (1 Mark) [Coordination Compounds] [PYQ 2014 / 2017] [Confidence: Very High]

Q33. (a) Describe the mechanism of the nucleophilic addition of Hydrogen Cyanide (HCN) to propanone to form a cyanohydrin. (2 Marks) (b) Explain the following named reactions with suitable balanced chemical equations: (i) Rosenmund reduction (ii) Hell-Volhard-Zelinsky (HVZ) reaction (2 Marks) (c) Account for the following: Chloroacetic acid is a significantly stronger acid than acetic acid. (1 Mark) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015 / 2021] [Confidence: Very High]

Answer Key

Q1. (b) Shows negative deviation from Raoult's law. (A maximum boiling azeotrope is formed due to stronger A-B interactions). **Q2.** (b) $\text{S cm}^2 \text{ mol}^{-1}$. **Q3.** (a) Zero. (The rate of photochemical reactions is often independent of the concentration of reactants). **Q4.** (c) Oxygen can form multiple bonds (double bonds) with transition metals. (This allows oxygen to stabilize higher oxidation states than fluorine, which can only form single bonds). **Q5.** (b) sp^3 . (Ni is in 0 oxidation state, $3d^8 4s^2$. Strong CO ligands force 4s electrons into 3d, making it $3d^{10}$. Empty 4s and 4p orbitals form sp^3 hybridization). **Q6.** (c) Pent-2-ene. (Follows Saytzeff's rule: the more highly substituted alkene is the major product). **Q7.** (c) Bromine water ($Br_2(aq)$). (Forms a white precipitate of 2,4,6-tribromophenol immediately). **Q8.** (b) Benzyl alcohol and Sodium benzoate. (Cannizzaro reaction due to absence of alpha-hydrogens). **Q9.** (d) 5. ($K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}$). **Q10.** (c) Negative. (According to the Debye-Hückel-Onsager equation, $\Lambda_m = \Lambda_m^\circ - A\sqrt{c}$). **Q11.** (b) Lower the activation energy. (It provides an alternative pathway). **Q12.** (d) $SOCl_2$. (Thionyl chloride produces gaseous by-products, SO_2 and HCl , leaving pure alkyl chloride). **Q13.** (a) Both A and R are true and R is the correct explanation of A. **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (a) Both A and R are true and R is the correct explanation of A. **Q16.** (d) A is false but R is true. (Oxidation of aldehydes is much easier than oxidation of ketones).

Q17. Raoult's law states that for a solution of volatile liquids, the partial vapour pressure of each component in the solution is directly proportional to its mole fraction in the solution ($P_A = P_A^\circ \cdot X_A$). Conditions for an ideal solution: $\Delta H_{mix} = 0$ and $\Delta V_{mix} = 0$. **Q18.** Faraday's First Law: The mass of a substance deposited or liberated at an electrode

during electrolysis is directly proportional to the quantity of electricity passed through the electrolyte ($w \propto Q$). Reduction of Al^{3+} : $Al^{3+} + 3e^- \rightarrow Al$. Therefore, 3 Faradays (3 F) of charge are required for 1 mole. **Q19.** (i) Transition elements show variable oxidation states because the energy difference between $(n-1)d$ and ns orbitals is very small, allowing electrons from both subshells to participate in bonding. (ii) Zinc ($3d^{10}4s^2$) has completely filled d-orbitals in its ground state as well as in its common +2 oxidation state, hence it is not considered a transition element. **Q20.** (i) Grignard reagents are highly reactive and strong bases; they instantly react with any source of protons (like water) to form alkanes ($RMgX + H_2O \rightarrow RH + Mg(OH)X$). (ii) Alkyl halides are insoluble because they cannot form hydrogen bonds with water, and the energy required to break the strong H-bonds in water is much greater than the new attractions formed.

Q21.

$3CH_3 - CH = CH_2 + \frac{1}{2}B_2H_6 \rightarrow (CH_3 - CH_2 - CH_2)_3B \xrightarrow{H_2O_2/OH^-} 3CH_3 - CH_2 - CH_2 - OH + B(OH)_3$. The major product is Propan-1-ol.

Q22. Isotonic solutions have the same osmotic pressure ($\pi_1 = \pi_2$). Since $\pi = CRT$, their molar concentrations must be equal: $C_1 = C_2$. $\frac{W_{urea}}{M_{urea} \times V} = \frac{W_{glucose}}{M_{glucose} \times V}$. $\frac{15}{60} = \frac{W_{glucose}}{180}$. $W_{glucose} = \frac{15 \times 180}{60} = 15 \times 3 = 45$ g. Mass of glucose is 45 g.

Q23. $t = \frac{2.303}{k} \log \left(\frac{[R]_0}{[R]} \right)$ $t = \frac{2.303}{1.15 \times 10^{-3}} \log \left(\frac{5}{3} \right) = \frac{2.303}{1.15 \times 10^{-3}} (\log 5 - \log 3)$

$t = \frac{2.303}{1.15 \times 10^{-3}} (0.6990 - 0.4771) = 2002.6 \times 0.2219 \approx 444.38$ seconds. **Q24.** Reaction: $Zn \rightarrow Zn^{2+} + 2e^-$ (Anode) and $Ag_2O + H_2O + 2e^- \rightarrow 2Ag + 2OH^-$ (Cathode). Number of electrons transferred, $n = 2$.

$\Delta G^\circ = -nFE^\circ_{cell} = -2 \times 96500 \text{ C mol}^{-1} \times 1.56 \text{ V} = -301,080 \text{ J mol}^{-1} = -301.08 \text{ kJ mol}^{-1}$. **Q25.** (i) The 5f electrons in actinoids have poorer shielding effect compared to the 4f electrons in lanthanoids, leading to a greater effective nuclear charge and stronger contraction. (ii) V_2O_5 acts as a catalyst because transition metals can easily change their oxidation states and form unstable intermediates. (iii) Cr^{2+} oxidizes to Cr^{3+} , gaining a stable exactly half-filled t_{2g}^3 orbital configuration in an aqueous medium. **Q26.** (i) Draw an octahedron with Co at the center. For *cis*, place two Cl^- on adjacent corners (90 degrees). For *trans*, place the two Cl^- on opposite corners (180 degrees). (ii) The *cis*-isomer has a non-zero dipole moment due to asymmetry. (iii) Linkage isomerism arises when an ambidentate ligand can bind through two different donor atoms. Example: $[Co(NH_3)_5(NO_2)]^{2+}$ (yellow) vs $[Co(NH_3)_5(ONO)]^{2+}$ (red).

Q27. (i) Phenol $+ NaOH \rightarrow$ Sodium phenoxide $\xrightarrow{CH_3I} Anisole + NaI$. (ii) Propan-2-ol $\xrightarrow{CrO_3} Propanone \xrightarrow{CH_3MgBr, H_3O^+} 2-Methylpropan-2-ol$. (iii) Cumene $\xrightarrow{O_2} Cumene hydroperoxide \xrightarrow{H^+, H_2O} Phenol + Acetone$.

Q28. (i) Propanone oxime: $(CH_3)_2C = N - OH$. (ii) Benzaldehyde phenylhydrazone: $C_6H_5 - CH = N - NH - C_6H_5$. (iii) Ethanal cyanohydrin: $CH_3 - CH(OH) - CN$.

Q29. (a) The time required for the concentration of a reactant to decrease to exactly half of its initial value. (b)

$t_{1/2} = \frac{0.693}{k}$. (c) Remaining $C - 14$ is 25%, which is $\frac{1}{4}$ of the original. This means it has undergone 2 half-lives ($100 \rightarrow 50 \rightarrow 25$). Age = $n \times t_{1/2} = 2 \times 5730 = 11,460$ years.

Q30. (a) 2,2-Bis(4-chlorophenyl)-1,1,1-trichloroethane (or p,p'-Dichlorodiphenyltrichloroethane). (b) In the stratosphere, UV radiation breaks the C-Cl bonds in CFCs, releasing highly reactive chlorine free radicals. These radicals catalyze the breakdown of thousands of ozone (O_3) molecules into oxygen (O_2), depleting the ozone layer. (c) Freon-12 formula: CCl_2F_2 . (Alternatively, Haloform reaction: $CH_3CH_2OH + 4Cl_2 + 6Ca(OH)_2 \rightarrow 2CHCl_3 + (HCOO)_2Ca + 5CaCl_2 + 6H_2O$).

Q31. (a) Molar conductivity (Λ_m) is the conducting power of all the ions produced by dissolving one mole of an electrolyte in a solution. For weak electrolytes, Λ_m increases steeply on dilution, especially at very low concentrations, due to a significant increase in the degree of dissociation (α) and the number of ions. (b) $c = 0.001028 \text{ mol L}^{-1}$;

$\kappa = 4.95 \times 10^{-5} \text{ S cm}^{-1}$. $\Lambda_m = \frac{\kappa \times 1000}{c} = \frac{4.95 \times 10^{-5} \times 1000}{0.001028} = 48.15 \text{ S cm}^2 \text{ mol}^{-1}$. Degree of dissociation

$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{48.15}{390.5} \approx 0.1233$. Dissociation constant

$K_a = \frac{c\alpha^2}{1-\alpha} = \frac{0.001028 \times (0.1233)^2}{1-0.1233} = \frac{1.56 \times 10^{-5}}{0.8767} \approx 1.78 \times 10^{-5} \text{ mol L}^{-1}$. **Q32.** (a) Cobalt (Z=27) is $3d^7 4s^2$. In $[CoF_6]^{3-}$, Co

is in +3 oxidation state ($3d^6$). Fluoride (F^-) is a weak field ligand, so pairing of 3d electrons does not occur. Outer 4d orbitals are used. Hybridization: sp^3d^2 (Outer orbital complex). Geometry: Octahedral. Magnetic property: Paramagnetic (4 unpaired electrons). (b) Diamminechloridonitrito-N-platinum(II). (c) Example: Hemoglobin, a coordination compound of Iron (Fe), is responsible for oxygen transport in blood. (Or Chlorophyll containing Mg, Vitamin B12 containing Co). **Q33.**

(a) Step 1: The nucleophile CN^- attacks the electrophilic carbonyl carbon of propanone, pushing the pi electrons onto the oxygen to form a tetrahedral alkoxide intermediate. Step 2: The intermediate captures a proton (H^+) from the reaction medium to form the cyanohydrin ($CH_3 - C(OH)(CN) - CH_3$). (b) (i) Rosenmund Reduction: An acid chloride is hydrogenated over a catalyst of palladium on barium sulfate to yield an aldehyde. Example:

$CH_3COCl + H_2 \xrightarrow{Pd-BaSO_4} CH_3CHO + HCl$. (ii) HVZ Reaction: Carboxylic acids with an alpha-hydrogen react with Cl_2 or Br_2 in the presence of red phosphorus to give alpha-halo carboxylic acids. Example:

$CH_3COOH + Cl_2 \xrightarrow{Red\ P, H_2O} Cl-CH_2-COOH + HCl$. (c) The highly electronegative chlorine atom in chloroacetic acid exerts a strong electron-withdrawing inductive effect (-I effect). This stabilizes the resulting carboxylate anion through charge delocalization, making it a stronger acid than acetic acid, which has an electron-donating methyl group (+I effect).

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.